

Report on HealthConnect Advanced Analytics & Decision Support

AnalystLab Africa Experience Lab -Week 6

1. Week 5 to Week 6 Transition

In Week 5, I worked with the HealthConnect appointment dataset and created a Power BI dashboard to better understand appointment attendance and no-show patterns. I also looked at key KPIs, patient behaviour and other appointment trends that could help explain why some patients miss their appointments.

One of the main things that stood out was the high no-show rate. Booking lead time and repeat no-show behaviour also looked important, but I felt they needed a deeper look before drawing stronger conclusions from them.

So, for Week 6, I'll be building on that work instead of starting over. I'll be taking a closer look at the strongest findings from Week 5, validating some of the important KPIs, improving the dashboard where necessary, and turning the findings into clearer recommendations for HealthConnect.

I'll also be working with someone on the Data Science track. I'll share some of my validated findings with her to see how they can support her no-show prediction model, and I'll also use her feedback to strengthen my own analysis and show how the two tracks connect.

2. Cross-Track Integration Readiness

- **Track I am collaborating with:** Data Science

i. What I need from the Data Science track:

I would like feedback on how useful my analytical findings are for the no-show prediction model, especially whether any of the patterns I identify can support feature refinement or model validation.

1

ii. What I will provide:

I will share one or more validated findings from my Week 6 analysis, such as a deeper breakdown of no-show behaviour by booking lead time, reminder status, appointment type, age group, or another useful segment.

iii. Why this connection is relevant:

The Data Science track is working on improving the no-show prediction model, while my role in Data Analytics is to understand the patterns behind no-show behaviour. My findings can help support modelling decisions, while feedback from the Data Science track can also help me understand which of my findings are most useful for prediction and decision-making.

iv. What I expect to improve through the collaboration:

I expect the collaboration to help me produce more useful and focused recommendations, while also giving the Data Science track evidence they can use when refining or validating their model.

v. Evidence I will keep:

I will keep the messages exchanged with my collaborator, the analytical finding I share, and her feedback on how the finding influenced her modelling decision.

That last part is important because the assignment says **communication alone is not enough**; the collaboration must lead to a meaningful change, decision, improvement or validated output.

3. Week 6 Analysis Focus

For Week 6, I decided to focus on some of the strongest findings from my Week 5 analysis rather than repeat the full exploratory analysis.

The main areas I will investigate further are booking lead time, repeat no-show behaviour, and other patient or appointment segments that may help explain no-show patterns more clearly.

In Week 5, booking lead time appeared to be an important factor, with no-show appointments having a longer average booking lead time than attended appointments. Repeat no-show behaviour also stood out, with patients who had previously missed appointments showing a high repeat no-show rate.

I will use the Week 6 analysis to check whether these patterns remain strong when broken down further and to identify any additional segment that could be useful for HealthConnect decision-making and for the Data Science no-show prediction model.

4. Advanced Analysis

4.1 Booking Lead Time Analysis

In Week 5, booking lead time was one of the things that stood out to me, so I decided to look at it more closely this week.

I grouped the appointments based on how many days in advance they were booked, then compared the attended appointments with the no-shows.

What I found was quite clear. For appointments booked within 0–7 days, the no-show rate was 29.47%. It went up to 35.19% for 8–14 days, 45.53% for 15–30 days, and 63.95% for appointments booked 31–60 days ahead.

From this, I can see that the longer the booking lead time, the higher the chance of a patient not showing up. This supports what I noticed in Week 5, but this time the pattern is even clearer.

I think this could be useful for HealthConnect because patients who book much earlier may need more follow-up or reminders as the appointment date gets closer.

For this particular analysis, I left out cancelled appointments and focused only on appointments that were either attended or recorded as no-shows.

4.2 Repeat No-Show Behaviour

Another area I wanted to look at more closely was whether patients who had missed appointments before were more likely to miss another one.

I grouped the patients based on their number of previous no-shows and compared this with their current appointment outcome.

The pattern was quite clear. Patients with no previous no-show had a no-show rate of 46.30%. This increased to 55.87% for patients with one previous no-show, 62.05% for those with two, and 70.33% for patients with three or more previous no-shows.

This shows that a patient's previous no-show history may be useful when identifying patients who are more likely to miss another appointment. For HealthConnect, this could help in deciding which patients may need more follow-up before their appointment date.

The group with three or more previous no-shows was much smaller than the other groups, so I would treat that result with some caution even though the overall pattern is still clear.

4.3 Additional Patient/Appointment Segment Analysis

I also wanted to check whether the type of appointment had any clear connection with no-show behaviour.

Follow-up appointments had the highest no-show rate at 54.17%, followed by diagnostic tests at 52.03%. Specialist consultations recorded 50.47%, while general consultations had the lowest rate at 49.14%.

Although there were some differences across the appointment types, the gap was not very wide. This tells me that appointment type may have some influence on no-show behaviour, but it does not appear to be as strong as booking lead time or a patient's previous no-show history.

For HealthConnect, follow-up appointments may still be worth paying closer attention to, but based on this analysis alone, I would not treat appointment type as one of the strongest indicators of a possible no-show.

5. KPI Validation and Refinement

5.1 Overall No-Show Rate

I first went back to check the overall no-show rate from Week 5. Out of the 5,000 appointments in the dataset, 2,423 were recorded as no-shows, which confirms the Week 5 no-show rate of 48.46%.

During the deeper Week 6 analysis, I also looked at the rate after removing cancelled appointments. When I compared only attended appointments and no-shows, the no-show rate was 51.15%.

I decided to keep both figures because they answer slightly different questions. The 48.46% gives the overall no-show rate across all appointments, while the 51.15% shows the no-show rate among appointments that ended either as attended or no-show.

5.2 Booking Lead Time KPI Validation

I also went back to check the average booking lead time from Week 5. The result still showed the same pattern. Patients who attended their appointments had an average booking lead time of 24.52 days, while those who did not show up had an average of 34.53 days.

This confirms what I found in Week 5, that longer booking lead time appears to be connected with a higher chance of a no-show. The deeper analysis I carried out this week also made the pattern clearer, especially when I broke the booking lead time into smaller groups.

Based on both results, I would still consider booking lead time an important KPI for HealthConnect when looking at no-show behaviour.

5.3 Reminder Effectiveness KPI Validation

I also checked whether sending reminders made any noticeable difference to appointment attendance.

For appointments where no reminder was sent, the no-show rate was 54.63%. For appointments where a reminder was sent, the no-show rate dropped to 49.86%.

For this comparison, I excluded cancelled appointments and focused only on attended appointments and no-shows. This is why these rates are slightly different from the reminder rates shown on the HealthConnect Overview dashboard, which includes all appointment outcomes.

This is a difference of about 4.77 percentage points, which supports the Week 5 finding that reminders may help reduce no-shows, even though the effect is not very large.

From this, I would still recommend reminders as part of HealthConnect's follow-up process, especially when combined with other risk factors such as long booking lead time or previous no-show history.

5.4 Refined KPI Summary

After validating the main Week 5 KPIs, the results still pointed in the same direction, but the Week 6 analysis gave a clearer picture.

The overall no-show rate remained 48.46% across all 5,000 appointments. When cancelled appointments were removed and only attended and no-show appointments were compared, the no-show rate was 51.15%.

Average booking lead time was 24.52 days for attended appointments and 34.53 days for no-shows. The grouped analysis also showed that no-show rates increased as booking lead time became longer.

Previous no-show history also showed a clear pattern, with the no-show rate increasing from 46.30% for patients with no previous no-show to 70.33% for patients with three or more previous no-shows.

Reminder status also remained useful, with appointments that received reminders showing a lower no-show rate of 49.86%, compared with 54.63% where no reminder was sent.

6. Dashboard Improvement

Instead of rebuilding my Week 5 dashboard, I kept the original dashboard as the HealthConnect Overview and added a second page for the deeper Week 6 analysis.

The new page focuses on the areas I investigated further this week. I added a visual showing how the no-show rate increases as booking lead time becomes longer, another showing how previous no-show history is connected with the chance of another missed appointment, and a third showing the differences across appointment types.

This made it easier to separate the general Week 5 overview from the more focused Week 6 findings without making the original dashboard too crowded.

HealthConnect Clinic Appointment - No-Show Analysis Dashboard

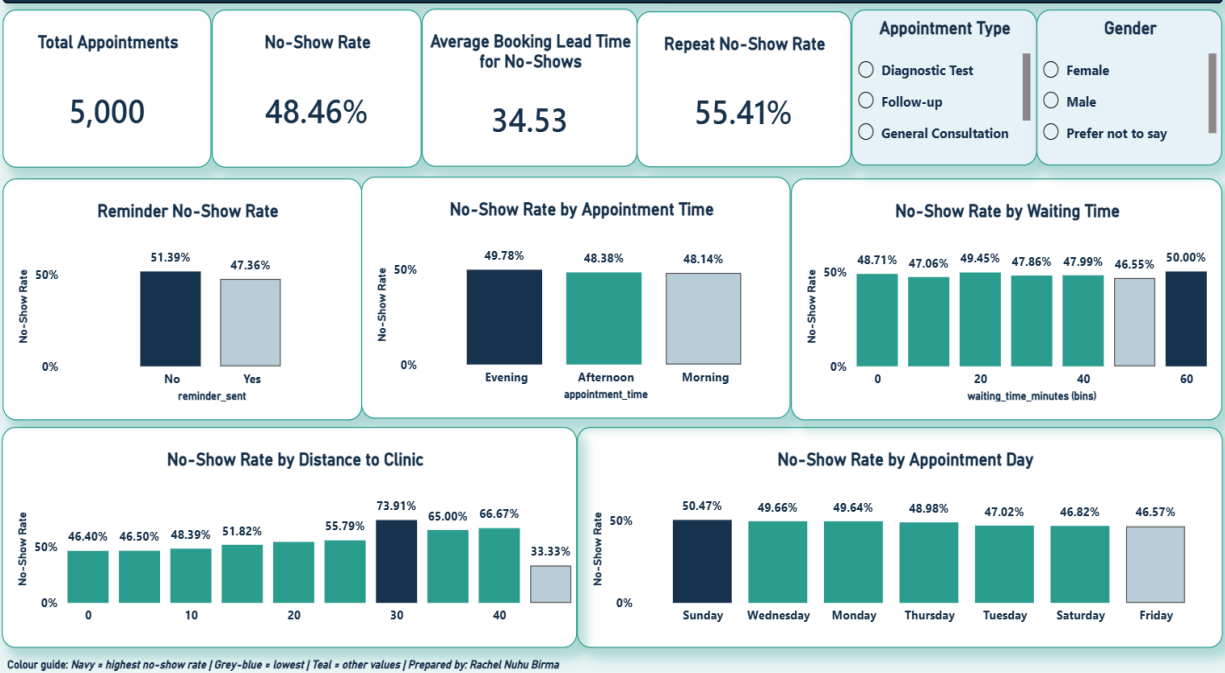


Figure 1: HealthConnect Overview Dashboard

HealthConnect Week 6 Advanced Analytics & Decision Support

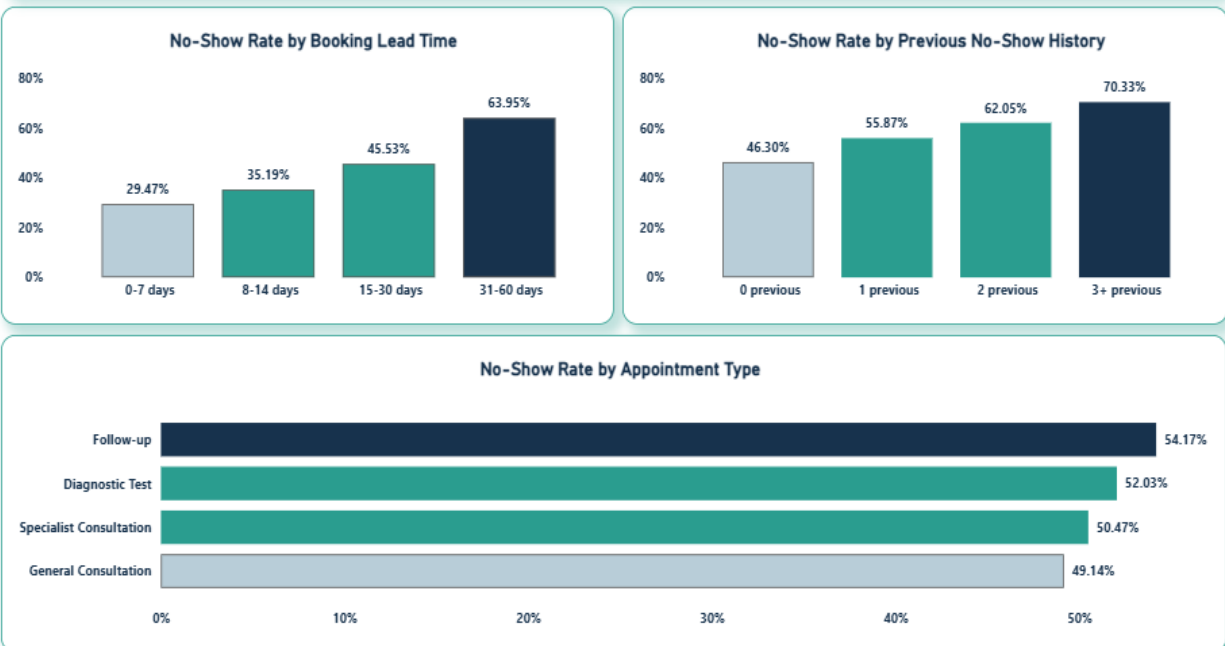


Figure 2: Week 6 Advanced Analysis Dashboard

7. Evidence-Based Insights and Business Recommendations

7.1 Booking Lead Time

One of the clearest findings from my Week 6 analysis was the steady increase in no-show rate as booking lead time became longer. The rate increased from 29.47% for appointments booked within 0 -7 days to 63.95% for appointments booked 31 - 60 days ahead.

Based on this, I think HealthConnect should pay closer attention to appointments booked far in advance. Instead of sending the same reminder to everyone, patients with longer booking lead times could receive an earlier reminder and another follow-up closer to their appointment date.

7.2 Previous No-Show History

Previous no-show history also stood out strongly. Patients with no previous no-show had a rate of 46.30%, while the rate increased to 55.87% after one previous no-show, 62.05% after two, and 70.33% for patients with three or more.

This suggests that previous attendance behaviour can help HealthConnect know which patients may need extra follow-up. Patients with a history of missed appointments could be placed on a priority reminder list or contacted more directly before their next appointment.

The group with three or more previous no-shows was much smaller, so I would still be careful about relying too heavily on that figure alone.

7.3 Appointment Type and Reminder Support

Appointment type showed smaller differences compared with booking lead time and previous no-show history. Follow-up appointments had the highest no-show rate at 54.17%, while general consultations had the lowest at 49.14%.

Reminders, however, showed a clearer difference. The no-show rate was 54.63% when no reminder was sent and 49.86% when a reminder was sent.

From this, I would recommend that HealthConnect continue using reminders, but make them more targeted. Patients with longer booking lead times, previous no-shows, or follow-up appointments may benefit more from additional reminder support than using the same approach for every patient.

8. Data Limitations and Remaining Issues

While the Week 6 analysis gave me a clearer picture of the no-show patterns, there are still some things I need to be careful about when interpreting the results.

First, some of the deeper analyses excluded cancelled appointments so that I could compare only attended appointments with no-shows. Because of this, those percentages should not be confused with the overall no-show rate of 48.46%, which was calculated using all 5,000 appointments.

I also noticed that some groups were much smaller than others. For example, the group with three or more previous no-shows had only 91 appointments. Although this group recorded the highest no-show rate, I would be careful about depending on that figure alone because the number of appointments in the group is relatively small.

Another limitation is that the analysis shows relationships between different factors and no-show behaviour, but it does not prove that any one of these factors directly causes a patient to miss an appointment.

Appointment type also showed only small differences in no-show rates, so I would not treat it as a strong indicator on its own.

One dependency I had during Week 6 was feedback from the Data Science track on the more detailed booking lead-time grouping. This was later tested independently, and the same no-show rates were confirmed. Although the refined grouping did not lead to a major improvement in model performance, it provided a more detailed and better validated version of the booking lead-time feature.

9. Cross-Track Integration Contribution

For my Week 6 cross-track collaboration, I worked with someone from the Data Science track who is improving the no-show prediction model. I shared my deeper booking lead-time analysis with her. My analysis grouped booking lead time into four ranges: 0 - 7 days, 8 - 14 days, 15 - 30 days, and 31 - 60 days.

One thing that became useful from the collaboration was that her earlier model had grouped booking lead time into only three categories: Short, Medium and Long. Her Medium group covered 8 - 30 days, but my analysis showed that this range actually had two different patterns. The no-show rate was 35.19% for 8 - 14 days and 45.53% for 15 - 30 days, which is about a 10 percentage-point difference. This meant that the wider 8 - 30-day group may have been hiding an important pattern. She explained that this could help her refine the booking lead-time feature in her model.

My collaborator independently recalculated the no-show rates using the four booking lead-time groups, and her results matched mine exactly: 29.47%, 35.19%, 45.53% and 63.95%. This gave us confidence that we were both working with the data consistently and that the refined grouping was correctly validated. She then tested the refined feature in both Logistic Regression and Random Forest models. The Logistic Regression results were almost unchanged from her Week 5 baseline, with the same accuracy, precision and recall to three decimal places. Random Forest showed a small improvement in accuracy and precision, but recall dropped slightly.

She decided to keep the refined booking lead-time feature because it was more detailed and better validated, but selected the refined Logistic Regression as her final model because it maintained recall. Since missing a likely no-show was considered more costly than a false alarm, recall was more important for that decision.

This helped me see that an analytical finding does not have to produce a large improvement in model performance to be useful. In this case, my analysis helped validate and refine a feature, while the modelling results also showed the limits of how much that feature could improve prediction on its own.

1. No-Show Rate by Booking Lead Time				
Count of appointment_id	Appointment Outcome			
Lead Time Group	Attended	No-Show	Grand Total	No-Show Rate
0-7 days	426	178	604	29.47%
8-14 days	372	202	574	35.19%
15-30 days	689	576	1,265	45.53%
31-60 days	827	1,467	2,294	63.95%
Grand Total	2,314	2,423	4,737	

2. No-Show Rate by Previous No-Show History				
Count of appointment_id	Appointment Outcome			
Previous No-Show Group	Attended	No-Show	Grand Total	No-Show Rate
0 previous	1,474	1,271	2,745	46.30%
1 previous	654	828	1,482	55.87%
2 previous	159	260	419	62.05%
3+ previous	27	64	91	70.33%
Grand Total	2,314	2,423	4,737	

3. No-Show Rate by Appointment Type				
Count of appointment_id	Appointment Outcome			
Appointment Type	Attended	No-Show	Grand Total	No-Show Rate
Diagnostic Test	272	295	567	52.03%
Follow-up	616	728	1,344	54.17%
General Consultation	1,007	973	1,980	49.14%
Specialist Consultation	419	427	846	50.47%
Grand Total	2,314	2,423	4,737	

4. Average Booking Lead Time by Appointment Outcome				
Appointment Outcome	Average of booking_lead_days			
Attended	24.52			
No-Show	34.53			
Grand Total	29.64			

5. No-Show Rate by Reminder Status				
Count of appointment_id	Appointment Outcome			
Reminder Sent	Attended	No-Show	Grand Total	No-Show Rate
No	583	702	1,285	54.63%
Yes	1,731	1,721	3,452	49.86%
Grand Total	2,314	2,423	4,737	

Figure 3: Week 6 Analysis Output Shared with Data Science

Search AnalystLab Africa

Onyinyechi Osuji

Messages

Add canvas

Monday, 7 September

Let me know what you've got whenever you're ready, no rush, but sooner is better sir

Rachel Birma

1:32 PM

Thanks Onyi for explaining it this way. I think this will work really well for our collaboration.

I'm still working through my Week 6 analysis, so I'll look closely at booking lead time and reminder status, but I'll also check other segments to see if anything more useful comes up.

Once I've validated the strongest finding, I'll definitely share that with you so you can see if it helps with your feature refinement or error analysis.

And yes, please share back what you find from the modelling side too. That will really help me document how our collaboration affected my own Week 6 work.

Thank you once again for your swift response.

3 replies

Last reply 20 hours ago

Yesterday

Rachel Birma

4:47 PM

Hello Onyi. I have one finding from my Week 6 analysis that might be useful for your model.

I looked into booking lead time, by grouping appointments into ranges and also comparing attended appointments with no-shows.

What I've found so far is that the no-show rates were:

- 0-7 days: 29.47%
- 8-14 days: 35.19%
- 15-30 days: 45.53%
- 31-60 days: 63.95%

So the pattern has become quite clear that the longer the booking lead time, the higher the no-show rate.

I excluded cancelled appointments from this particular analysis so that the comparison would only be between attended and no-show appointments.

And since booking lead time was already one of your stronger predictors, I thought this breakdown might help you decide whether grouping or transforming the feature could be useful to you. Let me know how you apply it on your side, because I'll also document that as part of my cross-track integration. Thank you. (edited)

Onyinyechi Osuji

11:49 PM

Thanks for this. For my booking_lead_category I used 3 bins: Short (0-7), Medium (8-30), Long (31+). The analysis you used 4 bins: 0-7, 8-14, 15-30, 31-60. The key insight from yours gives a more granular breakdown that shows that my "Medium" bin (8-30 days) was actually hiding two meaningfully different sub-patterns: 8-14 days sits at 35.2%, while 15-30 days jumps to 45.5%, a 10-point gap that my Week 5 3-bin version blurred together into a single "Medium" category.

Today

Rachel Birma

10:25 AM

This is really helpful. Please, if you decide to make any changes to the booking lead category from this, let me know what you changed and what you noticed after testing it. I'll also need that for my own collaboration evidence. Thank you.

Figure 4: Initial Data Science Collaboration and Feature Refinement Feedback

11

Project Track: Data Analytics | Prepared by: Rachel Nuhu Birma | 12th September, 2026

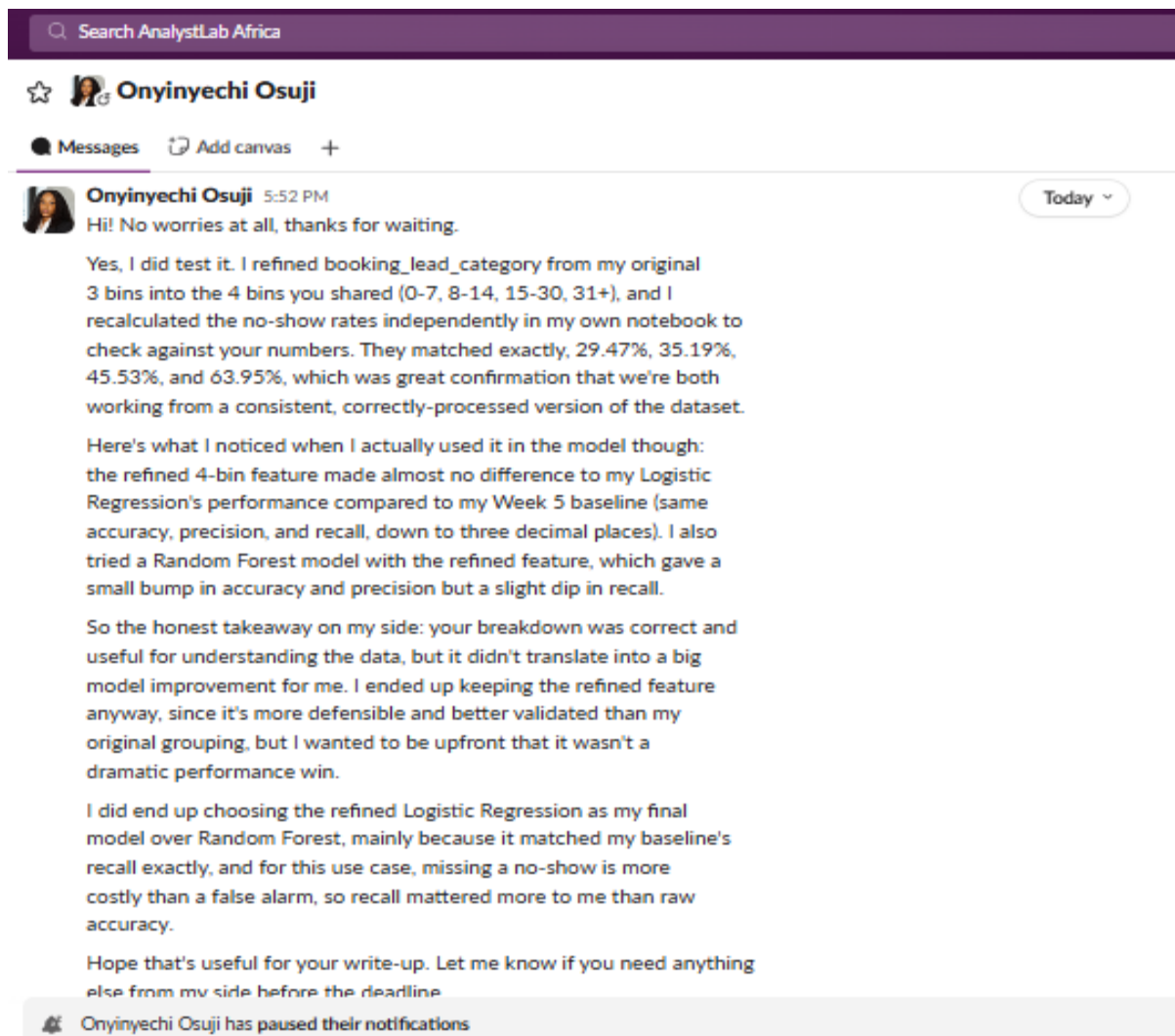


Figure 5: Data Science Validation, Model Testing and Final Modelling Decision

10. Week 7 Analytical Testing Requirements

Based on what I found this week, there are a few areas I would like to test further in Week 7 before treating the findings as final.

I would like to check whether the booking lead-time pattern remains consistent when combined with other factors such as reminder status and previous no-show history. For example, I want to see whether reminders make a bigger difference for patients whose appointments were booked further in advance.

I would also like to look more closely at previous no-show history, especially the smaller group of patients with three or more previous no-shows, to make sure the high rate I found is not being driven mainly by the small number of appointments in that group.

Another area I would like to test further in Week 7 is how the refined booking lead-time grouping behaves when combined with reminder status and previous no-show history. Since the Data Science testing showed that the refined grouping alone did not lead to a major improvement in model performance, I would like to see whether it becomes more useful when considered together with other risk factors.

I will also recheck the main Week 6 KPIs and dashboard calculations during Week 7 to make sure the figures remain consistent before the final project stage.

11. Week 6 Project Summary

This week, my main goal was to build on the work I completed in Week 5 by looking more deeply at the factors connected with missed appointments.

I focused mainly on booking lead time, previous no-show history, appointment type and reminder effectiveness. The deeper analysis showed that no-show rates increased as booking lead time became longer, and patients with a history of missed appointments were also more likely to miss another one.

I also validated some of the main KPIs from Week 5 and confirmed that the overall no-show rate remained 48.46%. The average booking lead time was also higher for no-shows than for attended appointments, and reminders were linked with a lower no-show rate.

I improved my Power BI dashboard by keeping the original HealthConnect overview and adding a second page for the Week 6 advanced analysis. This made it easier to show the deeper findings without making the original dashboard too crowded.

For my cross-track collaboration, I worked with someone from the Data Science track. I shared my more detailed booking lead-time analysis, which showed that the 8 - 30-day category she had previously used was hiding two different patterns. She independently validated my four-group analysis and got exactly the same no-show rates. She also tested the refined feature in her

Logistic Regression and Random Forest models. Although it did not produce a major improvement in model performance, she kept the refined grouping because it was more detailed and better validated. She also selected Logistic Regression as her final model because it maintained recall.

One of the main challenges for me this week was making sure that the deeper analysis remained consistent with the Week 5 KPIs and that the new dashboard calculations used the correct definitions.

In conclusion, my contribution this week was simply to turn some of the Week 5 findings into more detailed and useful insights that can support HealthConnect's decisions and also contribute to the Data Science modelling work.

My focus for Week 7 will be to test the findings further, especially the interaction between booking lead time, reminders and previous no-show history, and to see whether combining these factors gives clearer predictive or decision-making value.